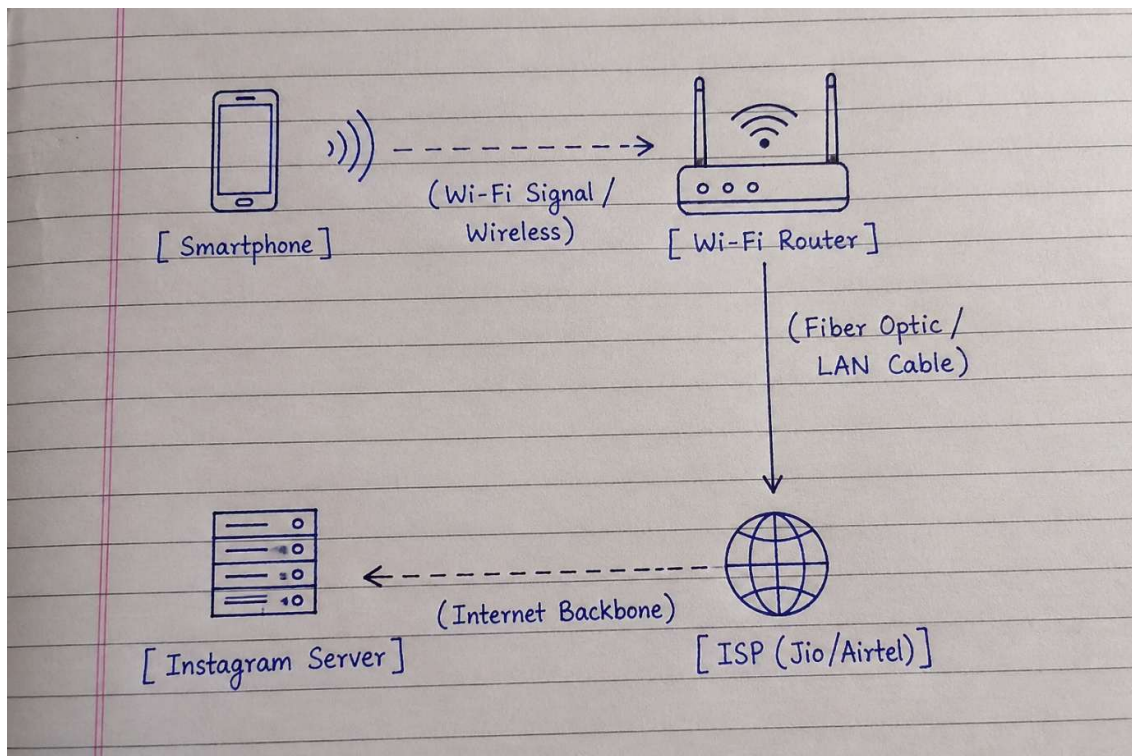


Task 1:

Network Connection Diagram (Phone to Instagram Server)

Objective: Understand how data travels from a local wireless device to a remote cloud server across local and global network infrastructure.



2. Component Details & Connection Labels:

1. Smartphone (End Device / Client):

- *Role:* Initiates requests (e.g., opening Instagram reels/feed) and sends data via 2.4GHz / 5GHz Wi-Fi radio signals.

2. Wi-Fi Connection (LAN Link):

- *Role:* Wireless medium connecting the smartphone to the local network router.

3. Wi-Fi Router (Local Access Point & Gateway):

- *Role:* Receives wireless packets, performs NAT (Network Address Translation), and routes local data to the external network via Ethernet.

4. ISP - Internet Service Provider (Jio / Airtel / Broadband):

- *Role:* Connects the local router to the regional network and routes traffic across high-speed submarine fiber-optic cables.

5. Internet Backbone (WAN Routing):

- *Role:* Global network of routers and switches directing traffic to Instagram's server IP address using public routing protocols.

6. Instagram Server (Data Center / Cloud Server):

- *Role:* Processes the incoming user request, fetches images/videos, and sends the requested data back through the same network path.

Task 2:

Daily Network Devices and Network Types

Objective: Identify 5 common network-connected devices used daily and specify the primary network type (PAN, LAN, WAN) associated with each.

1. Smartwatch

- **Network Type: PAN (Personal Area Network)**
- **Description:** Connects to a smartphone via Bluetooth over a short range (within 10 meters) to sync health data, notifications, and calls.

2. Laptop

- **Network Type: LAN (Local Area Network)**
- **Description:** Connects via Wi-Fi or Ethernet cable to a home/office router to share local resources (printers, files) and access local network services.

3. Smart TV

- **Network Type: LAN (Local Area Network)**
- **Description:** Connects to the home Wi-Fi network to communicate with local devices (screen casting) and route streaming requests through the home gateway.

4. Wi-Fi Router (Home Gateway)

- **Network Type: LAN & WAN (Local Area Network & Wide Area Network)**
- **Description:** Acts as the central hub creating the internal LAN for local devices while simultaneously connecting to the ISP's network to provide access to the global WAN (Internet).

5. Smartphone (on 4G/5G Mobile Data)

- **Network Type: WAN (Wide Area Network)**
- **Description:** Connects to distant cellular towers (telecom provider's infrastructure) to access internet services across vast geographical areas.

Task 3 :

Internet vs. Intranet Comparison

Objective: Differentiate between the Internet and an Intranet using a real-world educational institute / college network example.

1. Key Differences:

The Internet: A global, public network of interconnected computers accessible to anyone worldwide. It provides open access to public websites, services, and cloud platforms.

An Intranet: A private, restricted network accessible only to authorized users within a specific organization or campus. It is isolated from external public access using firewalls and internal IP routing.

2. Practical College / Public Wi-Fi Example:

Intranet Scenario (College Internal Network):

When connected to the College Wi-Fi, students can access the internal Student Portal / Library Server (e.g., accessing lab manuals, submitting assignments, or checking internal exam marks via 10.0.0.15 or portal.college.local).

This internal portal is not accessible from home or outside the campus because it is hosted exclusively on the college's private intranet.

Internet Scenario (Public Access):

While connected to the same Wi-Fi, using a web browser to search for information on Google, stream videos on YouTube, or access Wikipedia uses the Internet, routing traffic outside the local network to public global servers.

Task 4:

Data Flow in Mobile Application (Zomato Search Example)

Objective: Explain the end-to-end data transmission path from a mobile device to an application server and back during an in-app search query.

When I type "Pizza" in the **Zomato app** and hit search, my smartphone packages this search query into digital data packets and sends them over a local Wi-Fi or 4G/5G wireless network. The local router or cellular tower forwards these packets through the **Internet Service Provider (ISP)** network, which routes the data across the global Internet to Zomato's remote database server. The server processes the request, queries its database for matching nearby restaurants, and packages the results into a response. Finally, this data travels back through the ISP network to my phone, instantly displaying the restaurant search results on my screen.